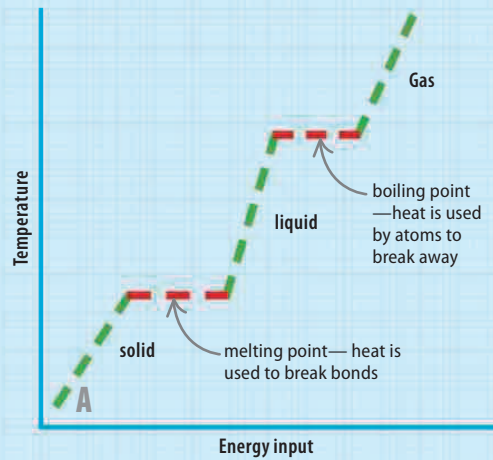


Latent heat

Energy cannot be created from nothing, nor can it be made to disappear. So when a substance is condensing or freezing, rearranging its units into a lower-energy state, the unneeded energy is given out, warming the surroundings as latent heat. The same amount of heat moves the other way, from the surroundings to the substance, when it is boiling or melting and moving to a more energetic state.

► Constant temperature

This graph shows that the temperature stays constant at the melting and boiling points (when the change of state is taking place). The increase or decrease of energy at these points is the latent heat.



Condensation

The reverse of boiling, condensation, occurs when gas molecules are unable to escape and form bonds with other molecules that pass close by. Gradually the molecules gather together into larger droplets of liquid.

Boiling

A liquid boils into a gas when it has enough energy to break all of its bonds. Instead of vibrating around a fixed point, the molecules of gas are free to move in any direction.

Melting

The vibrations in solids are too weak to break the bonds connecting them. The solid melts only when its units have enough energy to break a few of the bonds and become a liquid. Substances with high melting points have strong bonds connecting their units, and so need a lot of heat energy to break them.

Changing states in mixtures

Mixtures contain ingredients that have different melting and boiling points. When a solid is dissolved in a liquid, such as the salt in seawater, the mixture looks and behaves like a liquid. However, when it is heated to boiling point, the mixture separates—the water evaporates, leaving behind the solid salt (which melts at a much higher temperature).

nonmelting
chocolate chips

melting ice cream

► Melting mixtures

A chocolate chip ice cream is a mixture of ice, cream, and bits of chocolate. All of this is solid when the ice cream is served, but the ice and cream soon melt. The chocolate stays solid for longer, however.

